

Reproducible incident–reflected wave decomposition with embedded reliability diagnostics: WaveLabX, an open-source toolkit for laboratory wave-probe analysis

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Abstract

The separation of incident and reflected waves from co-linear probe arrays is a long-standing requirement of physical wave modelling, yet its implementations remain fragmented across laboratories and seldom report the reliability of the decomposition together with the result. This work presents WaveLabX, an open-source toolkit that unifies zero-crossing wave statistics with two- and three-probe frequency-domain decomposition under a single spectral formulation. Both two-probe Goda–Suzuki and three-probe redundant-array methods are implemented with the same Fourier coefficients, the same Goda probe-spacing band, and the same condition-number masking; the three-probe method additionally averages over admissible probe pairs and reports the fraction of measured spectral energy retained after filtering. Validation against synthetic records with prescribed reflection coefficients recovers the incident and reflected wave heights to within a few per cent across the spacing range $0.07 \leq \Delta x/L \leq 0.36$, and a sensitivity scan demonstrates that the three-probe routine remains within $\pm 5\%$ even when one pair sits outside the Goda band, because per-frequency masking discards the out-of-band pair. Application to a regular-wave and a JONSWAP-shaped irregular laboratory record reproduces physically admissible reflection coefficients with retained-energy fractions above 0.9. WaveLabX is distributed as a Python package and a zero-install client-side browser application; an automated JavaScript–Python cross-check confirms numerical agreement between the two interfaces. By attaching a quantitative measure of trust to every estimate, the toolkit aims to make

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reflection analysis in coastal-engineering laboratories more reproducible and more transparent.

Keywords: incident–reflected wave decomposition, wave reflection, Goda–Suzuki method, wave flume, physical modelling, coastal engineering

1. Introduction

Wave-probe measurements are central to the physical modelling of wave–structure interaction. Wave heights, spectra and reflection coefficients are required to assess coastal-protection systems, breakwaters and the quality of generated wave conditions in flumes and basins. Two analyses are almost universally needed: zero-crossing statistics of single-probe records, and the separation of a measured surface elevation into incident and reflected components.

The decomposition of incident and reflected waves from co-linear probe arrays is a long-established problem. The two-probe frequency-domain method of Goda and Suzuki [8] remains the most widely used approach, and least-squares multi-probe formulations [16, 10] improve robustness by adding redundancy. Although these methods are textbook material [7, 4, 9], their implementations are typically fragmented across laboratories as facility-specific scripts. This fragmentation limits reproducibility, cross-study comparability and the long-term preservation of experimental workflows.

A second, less widely appreciated issue motivates this work. Frequency-domain reflection analysis is sensitive to probe spacing and to the numerical conditioning of the inversion. The decomposition becomes singular when the probe spacing approaches an integer number of half wavelengths, and the classical guideline that the non-dimensional spacing satisfy $0.05 \leq \Delta x/L \leq 0.45$ bounds but does not eliminate this sensitivity. Even within the recommended band, certain probe configurations can yield ill-conditioned inversions and physically inconsistent estimates, and many existing implementations report a numeric result without an accompanying indication that it may be unreliable. WaveLabX addresses these two issues in a single transparent reference implementation. It packages zero-crossing statistics, the two-probe Goda–Suzuki method and a redundant three-probe array method, with the two decomposition methods built on one shared spectral formulation so that they are mutually consistent. Reliability diagnostics are embedded into the analysis: at every frequency the spacing guideline and the condition number of the inversion are evaluated, frequencies that fail either test are discarded, and the fraction of measured spectral energy retained after this filtering is reported. The user therefore obtains both incident and reflected wave quantities and

a quantitative basis for judging how much to trust them. Existing open tools cover parts of this workflow—the MATLAB zero-crossing routines of Neumeier [17] are one example, and other laboratory-specific implementations exist—but, to the authors’ knowledge, no openly licensed, documented and tested package currently unifies single-probe statistics with both two- and three-probe reflection analysis and exposes the underlying conditioning diagnostics in a single workflow. WaveLabX does not introduce a new reflection-decomposition theory; its contribution is a reproducible, open-source implementation that combines established two- and three-probe methods with automated reliability diagnostics and a no-install browser workflow.

The remainder of the manuscript is organised as follows. Section 2 describes the spectral formulation common to both decomposition methods, the reliability diagnostics and the implementation. Section 3 validates the methods against synthetic records with prescribed reflection coefficients and quantifies their sensitivity to probe spacing and gauge noise. Section 4 applies the three-probe routine to one regular and one irregular laboratory record from a wave flume. Section 5 discusses the implications of the embedded diagnostics for laboratory practice, and Section 6 concludes.

2. Methods

2.1. Linear wave theory and the spectral formulation

WaveLabX works in the framework of linear (Airy) wave theory. Wave kinematics are described by the dispersion relation

$$\omega^2 = g k \tanh(k h), \quad (1)$$

where $\omega = 2\pi/T$ is the angular frequency, T the wave period, $k = 2\pi/L$ the wavenumber, L the wavelength, h the still-water depth and g gravitational acceleration. For each frequency f at which a probe records, Eq. (1) is solved for k by Newton–Raphson iteration started from the explicit Fenton–McKee approximation [5], which converges reliably from deep to shallow water.

2.2. Two-probe Goda–Suzuki decomposition

The classical two-probe Goda–Suzuki frequency-domain decomposition [8] treats the measured surface elevation $\eta(x, t)$ as the superposition of an incident wave propagating in $+x$ and a reflected wave propagating in $-x$,

$$\eta(x, t) = \sum_n [a_{i,n} \cos(k_n x - \omega_n t + \phi_{i,n}) + a_{r,n} \cos(k_n x + \omega_n t + \phi_{r,n})]. \quad (2)$$

At every frequency ω_n the cosine and sine Fourier coefficients of the two probe records are combined to solve a 2×2 linear system for the incident

and reflected amplitudes, $a_{i,n}$ and $a_{r,n}$. A frequency is retained only if the non-dimensional probe spacing satisfies the Goda guideline,

$$0.05 \leq \Delta x/L \leq 0.45, \quad (3)$$

and the 2×2 inversion is acceptably conditioned. The incident and reflected spectra $S_i(f)$ and $S_r(f)$ are integrated to spectral wave heights and the reflection coefficient,

$$H_{m0} = 4\sqrt{m_0}, \quad m_0 = \int S(f) df, \quad K_r = \frac{H_{m0,r}}{H_{m0,i}}. \quad (4)$$

2.3. Three-probe redundant array decomposition

The three-probe routine uses three co-linear probes to form three independent probe pairs: (1–2), (1–3) and (2–3). The incident and reflected Fourier coefficients are computed for every pair using exactly the same formulation as the two-probe method; at each frequency the solutions from the pairs that satisfy Eq. (3) and the conditioning test are averaged. This redundancy improves robustness when one pair is unfavourably spaced, and is closely related to the least-squares multi-probe estimators of Mansard and Funke [16] and Kobayashi et al. [10], specialised to the three-probe geometry that is in widespread laboratory use.

2.4. Reliability diagnostics

Both decomposition routines compute the condition number of the 2×2 inversion at every frequency and discard frequencies above a configurable threshold. Both report a *retained-energy fraction*, defined as the fraction of measured spectral energy that survives the spacing and conditioning filters. The three-probe routine warns the user when this fraction falls below a threshold (default 0.8); this provides an explicit, quantitative criterion for when a decomposition should be treated as unreliable, and answers the practical question of how to interpret a result for which a large part of the spectrum has been rejected. This diagnostic is, to the authors' knowledge, the principal practical addition of WaveLabX over textbook implementations of the Goda–Suzuki method.

2.5. Implementation

WaveLabX is implemented as a small Python package, `wavelabx`, accompanied by a self-contained browser application in the `web/` directory. The package is deliberately modular: `core` contains the dispersion solver; `stats` contains the zero-crossing routines; `two_probe` and `three_probe` implement the two decomposition methods on the common spectral formulation of Section 2;

`analysis` provides a high-level workflow that selects a recommended method; and `sensitivity` contains the known-truth synthetic-record generators used in Section 3.

The two-probe and three-probe methods are *alternatives*, not a procedural sequence: WaveLabX evaluates the probe geometry and selects the method expected to be more reliable for the given array. The three-probe method is preferred when it retains a sufficient fraction of the measured spectral energy, and a well-spaced two-probe pair is used otherwise. The two methods are exposed as separate entry points in both interfaces: `two_probe_goda` and `three_probe_array` in the Python package, and the corresponding `twoProbeGoda` and `threeProbeArray` routines in `web/spectral.js`. A minimal Python example is:

```
import numpy as np
from wavelabx import reflection_analysis

# eta: (N, 3) array of probe elevations [m]
eta = np.loadtxt("data/wavedata.csv", delimiter=",")

# fs in Hz, depth h in m, gpos in m
out = reflection_analysis(
    eta, fs=100.0, h=0.25, gpos=(0.0, 0.35, 0.70))

tp = out["three_probe"]
print(out["method_used"], tp["Kr"],
      tp["retained_energy_fraction"])
```

The browser application (Fig. 2) is hosted at <https://wave-lab-x.vercel.app>; no installation or server is required and no data leaves the user's machine. The numerical core, `web/spectral.js`, ports the Python entry points to JavaScript using a Bluestein [2] discrete Fourier transform that handles records of arbitrary length. The CSV column count selects the routine: two columns invoke the two-probe Goda–Suzuki method; three columns invoke the redundant three-probe array; six columns process two co-linear three-probe arrays. Three automated JavaScript–Python cross-checks in the test suite confirm that the two interfaces agree numerically on identical inputs to several decimal places.

3. Validation

3.1. Recovery of a known synthetic truth

Because incident and reflected components cannot be measured independently in a physical flume, WaveLabX is validated against synthetic three-gauge

**WaveLabX — Software Architecture
for Incident–Reflected Wave Decomposition**

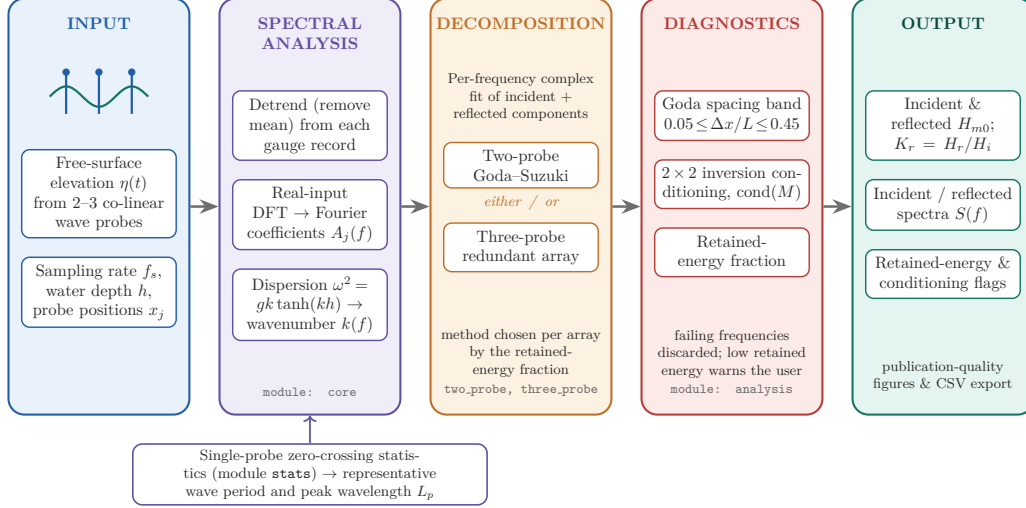


Figure 1: WaveLabX software architecture. The analysis pipeline runs from left to right; the two-probe and three-probe methods are a choice, not a sequence, selected per array from the probe geometry and the fraction of spectral energy the three-probe method retains. Both the Python package and the browser application implement this identical pipeline.

records generated from linear wave theory with prescribed incident height, reflection coefficient and probe geometry (`wavelabx.sensitivity`). For a well-spaced array, both methods recover the incident and reflected spectral wave heights and the reflection coefficient to within a few per cent of the prescribed truth (Fig. 3). The decomposition is therefore shown to be correct in a controlled setting before it is applied to laboratory data.

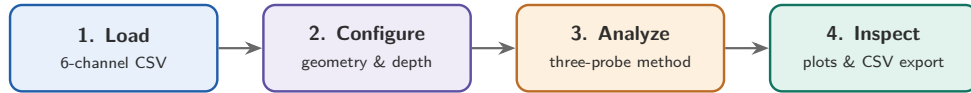
3.2. Probe-spacing sensitivity

To quantify how probe spacing controls accuracy, both methods were applied to known-truth synthetic records under deliberately different geometric conditions (Fig. 4); the two panels are designed to highlight the distinct purpose of each method.

Panel (a) is a single-pair two-probe scan: at every $\Delta x/L$ the gauges sit at $(0, \Delta x, 2\Delta x)$ and only the first pair is used. The recovery stays within $\pm 5\%$ for approximately $0.07 \leq \Delta x/L \leq 0.36$, with one borderline point at $\Delta x/L \approx 0.14$ reaching $\approx 5\%$; outside this range the error degrades smoothly because the inversion approaches a singularity.

Panel (b) tests the redundancy of the three-probe routine on an *asymmetric* geometry: the first spacing is held fixed at $X_{12} = 0.10 L$ (well inside the

The WaveLabX Browser Application — Client-Side Reflection Analysis



(a) Configuration: probe arrays, sampling rate and water depth

Settings

Sampling freq (Hz): 100 Detect band min (Hz): 0.10 Detect band max (Hz): 4.00 Skip first (waves): 0 Analyze (waves, 0 = all): 0 Water depth d (m) — all files: 0.25 [Apply to all](#)

Period & wavelength [?](#) Analysis method [?](#)
 Tm (zero-crossing mean) Auto (3-probe + 2-probe fallback)

Array 1 — ch 1–3 (seaward)
 X 12 0.45 X 23 0.30

Array 2 — ch 4–6 (shoreward)
 X 45 0.30 X 56 0.45

Each array is defined by two gauge spacings; gauge positions are taken as 0, X_{12} , $X_{12}+X_{23}$. Water depth and gauge layout apply to every file; wave frequency is detected per file and is editable in the table. The number of gauges per file is auto-detected from the CSV: 2-column CSVs run two-probe Goda-Suzuki on the single pair (only X_{12} is used); 3-column CSVs run three-probe on Array 1 only; 6-column CSVs run two three-probe arrays.

(b) Time series



(c) Decomposed spectrum



(d) Per-probe spectrum



(e) Per-file results: incident/reflected heights and coefficients

Results · 1 file(s)										
File	d (m)	f_p (Hz)	T_p (s)	H_{I1} (m)	H_{R1} (m)	K_{r1}	H_{I2} (m)	H_{R2} (m)	K_{r2}	K_t
jonswap_example.csv	0.5	0.785	1.274	0.0557	0.0447	0.803	0.0315	0.0570	1.810	0.566

Figure 2: The WaveLabX browser application, applied to a six-gauge laboratory record. The tool follows a four-step workflow: load, configure, analyse, inspect. (a) The configuration panel sets the two probe arrays, the sampling rate and the water depth. (b–d) The interactive visualisation offers time-series, decomposed incident/reflected and raw per-probe spectrum views, each with zoom, pan and per-point readout. (e) The results table reports the incident and reflected wave heights and the reflection coefficients of both arrays for every file, and exports them as CSV. The application runs entirely in the browser, with no installation and no server.

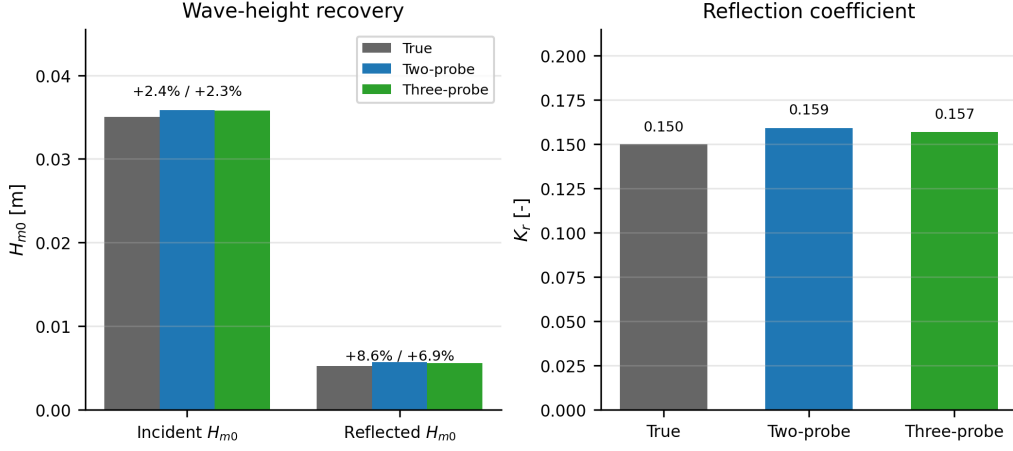


Figure 3: Recovery of incident/reflected wave heights and the reflection coefficient against a known synthetic truth, for a well-spaced array. Both methods agree with the prescribed values to within a few per cent.

Goda band) while the second spacing X_{23} is varied across the same range as panel (a). The routine stays within $\pm 5\%$ across virtually the entire varied range, even where X_{23} is far outside the Goda band, because the per-frequency masking keeps the in-band pairs and discards the out-of-band one. No single-pair two-probe analysis can match this behaviour; this is the redundancy benefit the three-probe method was designed for.

Both scans were repeated at three noise levels (3%, 10% and 30% of the incident height), covering typical flume signal-to-noise ratios and a deliberately extreme stress test. Inside the recommended spacing band the noisy two-probe curves are visually indistinguishable from the noise-free curve up to 10% noise; at 30% noise a small but visible spread appears. The three-probe panel shows almost no noise sensitivity at any level, because the in-band pair carries the estimate. The default retained-energy threshold of 0.8 separates the in-band, $\pm 5\%$ regime from the regime in which the routine flags the case automatically: at $\Delta x/L \approx 0.04$ in the equal-spacing two-probe case, for example, the retained energy falls below 0.4 and the result is flagged as unreliable.

3.3. Robustness on non-uniform geometries

Beyond the varied asymmetric geometry of Fig. 4(b), three representative non-uniform geometries illustrate the redundancy benefit and the failure mode of the three-probe routine. A well-spaced array (0, 0.35, 0.70) keeps all three pairs inside the band and recovers the incident height to within 3.3%. An array of mixed spacing (0, 0.10, 0.60) keeps all pairs admissible and recovers

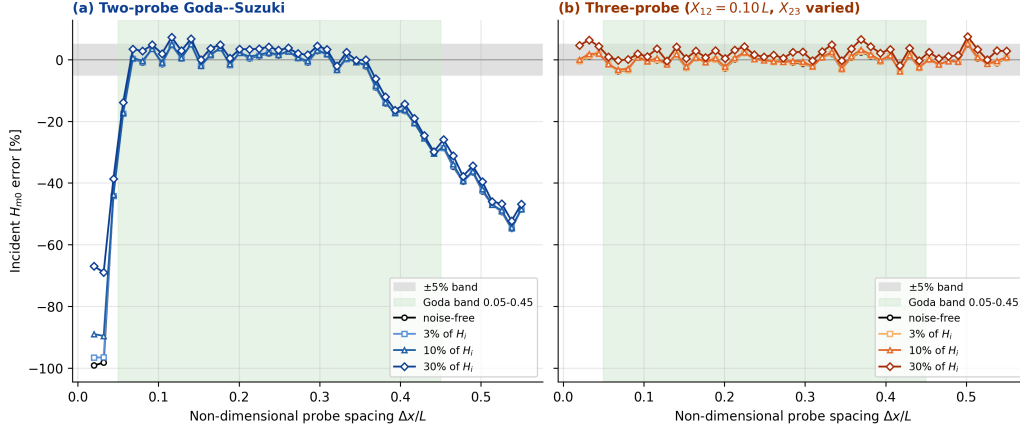


Figure 4: Probe-spacing sensitivity. Panel (a): two-probe scan with $X_{12} = \Delta x$. Panel (b): three-probe scan with the first spacing held fixed at $X_{12} = 0.10L$ (inside the Goda band) and the second spacing $X_{23} = \Delta x$ varied across the same range. Each panel is shown at four gauge-noise levels (noise-free, and 3%, 10% and 30% of the true incident wave height). The shaded grey band marks $\pm 5\%$ incident-height error; the green band marks the classical Goda spacing range $0.05 \leq \Delta x/L \leq 0.45$. The three-probe panel stays inside the $\pm 5\%$ band over the entire varied range, showing that the routine’s per-frequency pair masking lets it recover the truth even when one pair is well out of the Goda band.

the truth to within 3.1%. An array in which all probes are too closely spaced (0, 0.05, 0.10) leaves only one pair marginally admissible: the retained-energy fraction collapses to 0.58 and the incident height is in error by roughly 22%, which the routine flags through the retained-energy diagnostic. The default threshold of 0.8 is therefore not arbitrary; it is the value below which the $\Delta x/L \in [0.07, 0.36]$ guarantee of Fig. 4 breaks down because no admissible pair remains.

4. Application to laboratory datasets

To exercise WaveLabX across regular, irregular and known-truth scenarios, the three-probe method was applied to three contrasting cases (Table 1), all reproduced by the script `multi_example_test.py` in the `scripts/` directory:

1. *Synthetic regular wave with prescribed reflection*. A monochromatic incident–reflected superposition is generated from linear theory ($T_p = 1.25$ s, $h = 0.50$ m, $H_i = 0.113$ m, prescribed $K_r = 0.300$) and the three-probe routine is asked to recover the truth.
2. *Real regular wave* ($f = 1.25$ Hz, H_3 target, $h = 0.35$ m). Recorded as six-channel data at 100 Hz in `regular_example.csv` under `data/`; the seaward three-gauge array (channels 1–3) is analysed with the spacings of Fig. 5, $X_{12} = 0.60$ m and $X_{23} = 0.30$ m.

3. *Real irregular wave (JONSWAP, $h = 0.50$ m).* Recorded as six-channel data at 100 Hz in `jonswap_example.csv` under `data/`; the seaward array is analysed with the browser-application default spacings $X_{12} = 0.45$ m and $X_{23} = 0.30$ m.

Fig. 5 shows the wave-flume layout used to acquire the two laboratory records: a 15 m flume with a piston wavemaker at the seaward end and an absorbing beach at the shoreward end, an instrumented submerged structure near mid-flume, and two co-linear three-gauge arrays (channels P1–P3 seaward and P4–P6 shoreward) placed between the wavemaker, the structure and the beach.

Table 1 summarises the results. The synthetic case recovers the prescribed $K_r = 0.300$ to three decimal places, the real regular wave is recovered with a clean physical $K_r = 0.13$ and retained-energy fraction 0.94, and the JONSWAP array returns a physically admissible $K_r = 0.80$ with retained-energy fraction 0.95. Fig. 6 shows the JONSWAP incident, reflected and composite spectra.

Case	Type	Data file	Spacings (m)	H_i (m)	H_r (m)	K_r	Retained
Synthetic regular	regular	<code>wavelabx.sensitivity</code>	0.60, 0.30	0.1131	0.0339	0.300	1.00
Regular wave (real)	regular	<code>regular_example.csv</code>	0.60, 0.30	0.1199	0.0151	0.126	0.94
JONSWAP (real)	irregular	<code>jonswap_example.csv</code>	0.45, 0.30	0.0557	0.0447	0.803	0.95

Table 1: Summary of the three illustrative cases analysed in Section 4, the data file used and the seaward-array gauge spacings X_{12}, X_{23} of each. For the synthetic case a prescribed reflection coefficient $K_r = 0.300$ was imposed and recovered exactly. The regular-wave case uses the spacings shown in Fig. 5; the JONSWAP case uses the Wave-LabX browser-application default spacings. Data files are located under `data/` in the repository; numbers are reproduced by `scripts/multi_example_test.py` and stored in `results/multi_example_results.csv`.

The regular-wave case illustrates the diagnostic in action. At $f = 1.25$ Hz and $h = 0.35$ m the wavelength is $L \approx 0.98$ m, so the three pairs sit at $\Delta x/L \approx 0.61, 0.92$ and 0.31 ; only pair (2–3) lies inside the Goda band $0.05 \leq \Delta x/L \leq 0.45$. Per-frequency masking inside `three_probe_array` reduces the calculation at the dominant frequency to a single-pair estimate on pair (2–3), and the retained-energy fraction of 0.94 confirms that nearly all spectral energy is admissible. That the synthetic case recovers $K_r = 0.300$ exactly under the same conditions shows the single-pair fallback to be self-consistent. The per-pair Goda warning emitted by both the Python package and the browser application makes this state visible at a glance.

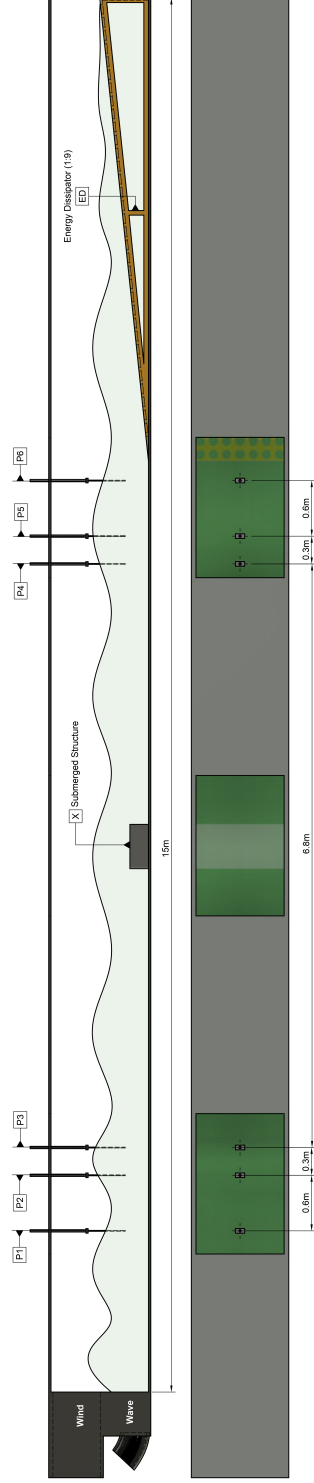


Figure 5: Wave-flume experimental setup that produced the laboratory records analysed in Section 4. The campaign is described in Lamsal et al. [15]. Top: elevation view, showing the wavemaker (left), the two co-linear three-gauge probe arrays P1–P3 and P4–P6, an instrumented submerged structure near mid-flume, and the absorbing beach at the shoreward end. Bottom: plan view, showing the in-flume location of each three-gauge array and the submerged structure with measured spacings.

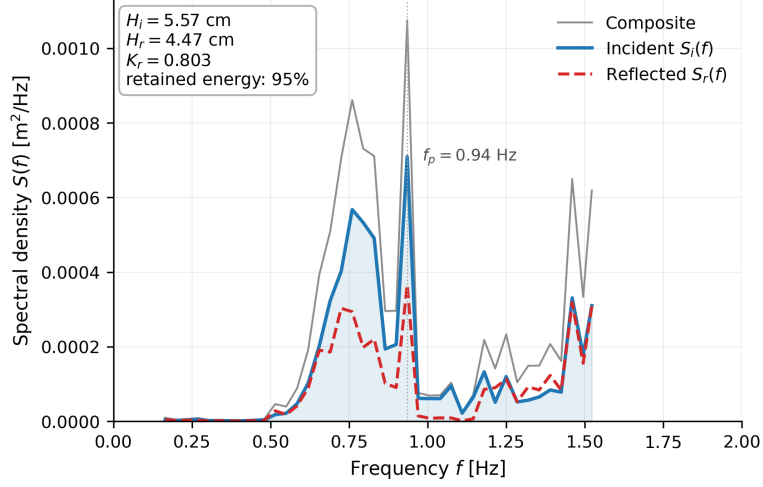


Figure 6: Incident, reflected and composite wave spectra from the real irregular-wave (JONSWAP) laboratory record `data/jonswap_example.csv`, computed for the seaward three-gauge array (channels 1–3).

5. Discussion

The validation and application results lead to three observations of practical relevance for laboratory wave analysis.

The Goda band is a necessary but not sufficient condition.. Within the textbook band $0.05 \leq \Delta x/L \leq 0.45$ the two-probe inversion can still be ill-conditioned at specific frequencies, and a single-pair analysis can drift outside $\pm 5\%$ near the band edges even in the absence of noise (Fig. 4a). A practical implementation that returns only the inverted coefficients, without an explicit conditioning test, therefore embeds an unflagged source of error. The condition-number filter applied at every frequency in WaveLabX, together with the retained-energy fraction reported with the estimate, makes this source of error observable.

The three-probe method earns its complexity through its diagnostics.. If all three pairs of a three-probe array satisfy the Goda band, the three-probe method is a modest accuracy improvement over the best two-probe pair. Its decisive advantage emerges when the array is non-uniform, as is common in practice, where the per-pair masking inside `three_probe_array` keeps the in-band pairs and discards the out-of-band ones automatically (Fig. 4b). The cost is the bookkeeping of three 2×2 inversions per frequency; the benefit is a result with a self-reported retained-energy fraction, that is, a quantitative

estimate of the spectral fraction on which the decomposition was actually performed.

The retained-energy threshold is a design-time tool.. The spacing-sensitivity and non-uniform-geometry results indicate that the default threshold of 0.8 is the value below which the $\Delta x/L \in [0.07, 0.36]$ guarantee of Fig. 4 breaks down because no admissible pair remains. Experimentalists can use this criterion not only after the test, to flag an unreliable estimate, but also before the test, when choosing probe positions: running the spacing scan in `wavelabx.sensitivity` for the target period range identifies geometries that will return retained-energy fractions above 0.8 across the spectrum, and is a practical alternative to repeatedly re-running an entire campaign with suboptimally placed probes.

The software was developed alongside laboratory programs at the SUSTAIN Laboratory of the University of Miami and has supported a body of experimental wave–structure-interaction research, including studies of submerged SEAHIVE breakwaters [15], perforated breakwater structures [13], hybrid coral-reef structures [12], wave dissipation by mangrove forests [3], hurricane-induced wind and wave loading [1], beach-erosion-control modelling and design [6], and a doctoral dissertation on submerged breakwaters for coastal protection [11]; the toolkit itself is openly archived [14].

WaveLabX targets one-dimensional wave propagation with co-linear probe arrays and linear-wave behaviour; strongly nonlinear waves, breaking and multi-directional wave fields are outside its current scope. The embedded diagnostics also rely on the linear superposition assumed by the Goda–Suzuki and three-probe formulations; their reliability under strongly nonlinear conditions is a topic for future work.

6. Conclusions

This work presented WaveLabX, an open-source toolkit that unifies zero-crossing wave statistics with two- and three-probe incident–reflected decomposition under one spectral formulation. Reliability diagnostics—probe-spacing checks, inversion conditioning and a retained-energy fraction—are embedded directly in the analysis, attaching a quantitative measure of trust to every estimate.

Validation against synthetic records with prescribed reflection coefficients showed that both methods recover incident and reflected wave heights to within a few per cent for well-conditioned arrays; that the two-probe error grows predictably outside the spacing range $0.07 \leq \Delta x/L \leq 0.36$; and that the three-probe method’s per-frequency masking keeps the estimate within

$\pm 5\%$ even when one pair is far outside the Goda band, because the masking discards that pair automatically. Application to laboratory records of one regular wave and one JONSWAP-shaped irregular wave produced physically admissible reflection coefficients with retained-energy fractions above 0.9, and demonstrated that the per-pair Goda diagnostic correctly identified the configurations in which the estimate reduced to a single admissible pair. Distributed as a Python package and a zero-install browser application—the two interfaces sharing an automated numerical cross-check—WaveLabX is intended to support reproducible reflection analysis across coastal- and hydraulic-engineering laboratories. Future work includes extending the methods to oblique and multi-directional wave fields.

Data and code availability

The WaveLabX source code, example datasets, reproducible scripts and the browser application are openly available under the MIT License at <https://github.com/sandslamsal/WaveLabX>. The version that supports the figures, tables and analyses reported in this manuscript is archived on Zenodo as Lamsal et al. [14] (doi:10.5281/zenodo.20217994). The example records analysed in Section 4 (`regular_example.csv` and `jonswap_example.csv` under the `data/` directory) are bundled with the release. Running `multi_example_test.py` under `scripts/` regenerates the numerical results in Table 1; running `real_data_example.py` regenerates the spectra in Fig. 6. The browser application is deployed at <https://wave-lab-x.vercel.app> and runs the same analysis routine as the Python package.

Acknowledgements

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